

Final Project Report

Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau

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Team Member 1: **POLEPALLI VENKATA JAGADEESH**

. INTRODUCTION

Project Overview

The increasing demand for safe and reliable transportation infrastructure highlights the need for smarter bridge monitoring solutions. This project aims to develop a Smart Bridge system that organizes and analyzes bridge-related data to understand performance, usage, and potential risks.

Purpose

The purpose of this project is to develop a Smart Bridge system that helps in monitoring bridge performance, analyzing data, and presenting insights in a clear and visual manner. It aims to improve safety, support timely maintenance, and enable better decision-making through a structured project approach.

This project aims to:

This project aims to:

1. Analyze electricity consumption patterns across different states and regions in India.
2. Compare usage trends over time (year-wise and month-wise) to identify growth or decline.

3. Identify top and bottom electricity-consuming states and regions.
4. Visualize regional and temporal variations using interactive Tableau dashboards.
5. Support better decision-making for energy planning, management, and sustainability.

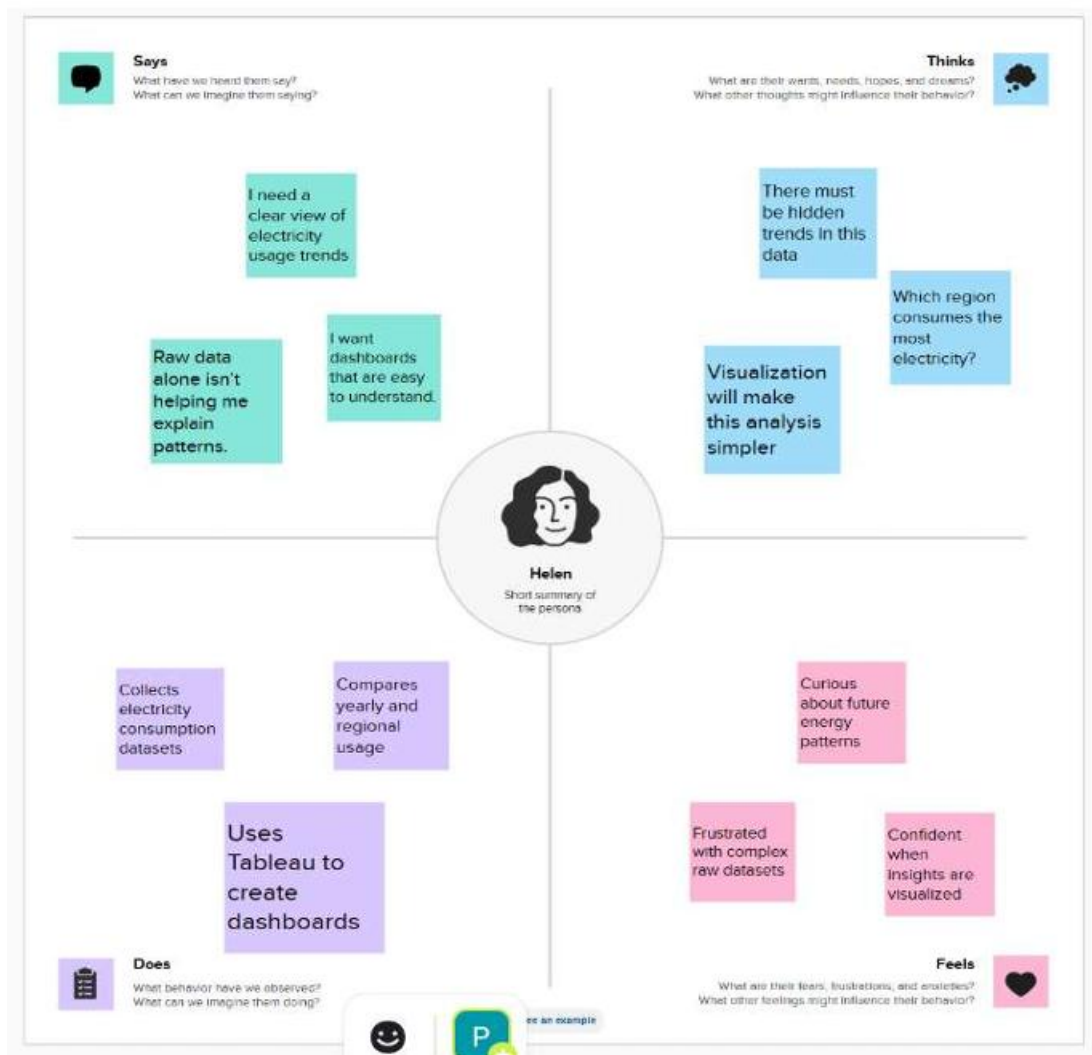
1. IDEATION PHASE

1.1 Problem Statement



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A data analyst	Analyze electricity consumption patterns across regions and years	The data is large and difficult to interpret	Raw data does not clearly show trends or variations	The need for interactive visual dashboards
PS-2	A decision maker / policy analyst	Identify high and low electricity consumption areas	Comparisons across regions are time-consuming	Consumption varies significantly over time and location	Visualization is essential for better planning

1.3 Empathy Map



1.2 Brainstorm



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

🕒 10 minutes to prepare
🕒 1 hour to collaborate
👤 2-8 people recommended

➕

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 10 minutes

➕

Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

➕

Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

➕

Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#)

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

How might we [your problem statement]?



Key rules of brainstorming

To run an smooth and productive session

- Stay in topic.
- Encourage wild ideas.
- Defer judgment.
- Listen to others.
- Go for volume.
- If possible, be visual.

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

TIP

You can select a sticky note and hit the pencil (switch to sketch) icon to start drawing!

Person 1

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 2

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 3

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 4

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 5

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 6

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

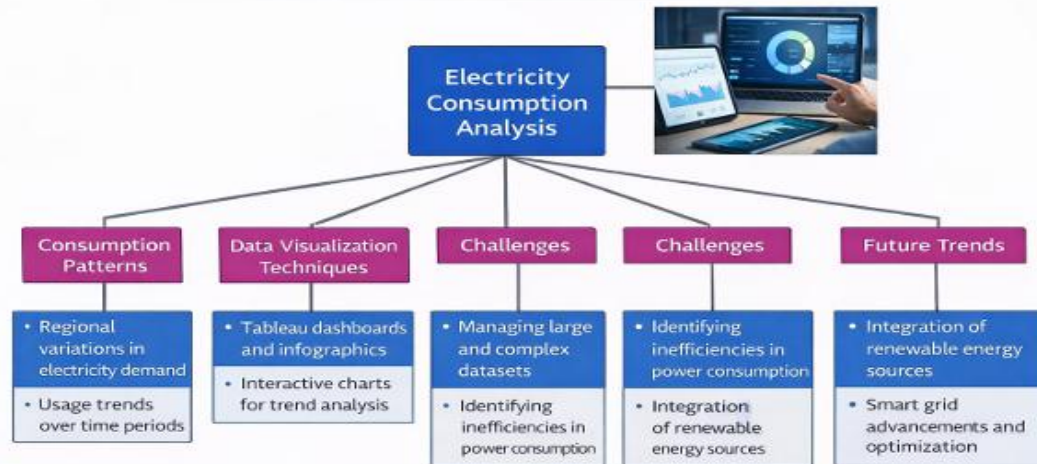
Person 7

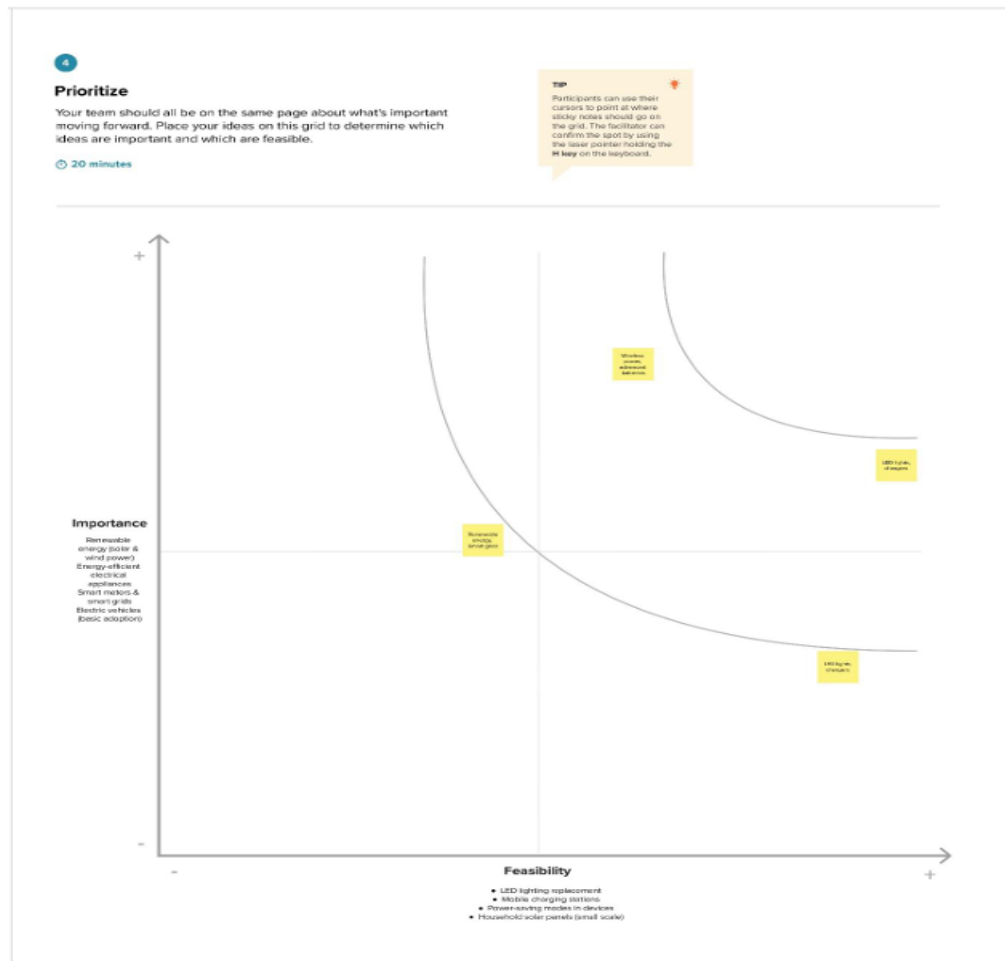
Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Person 8

Reducing household energy usage
Developing a smart home system
Reducing household energy usage

Electricity Consumption Analysis





2.REQUIREMENT ANALYSIS

2.1 Customer Journey map



2.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion	Load electricity consumption data from CSV/Excel files
FR-2	Data Preprocessing	Support multiple datasets (year-wise / region-wise data)
FR-3	Data Analysis	Analyze electricity consumption by year (2019, 2020)
FR-4	Visualization	Create charts (bar, line, pie, map) for consumption analysis
FR-5	Interactive Dashboard	Provide filters for year, state, and region

FR-6	Comparative Analysis	Compare electricity consumption across different years
FR-7	Storytelling	Create a Tableau story for step-by-step insight explanation
FR-8	Reporting & Insights	Summarize key findings from dashboards

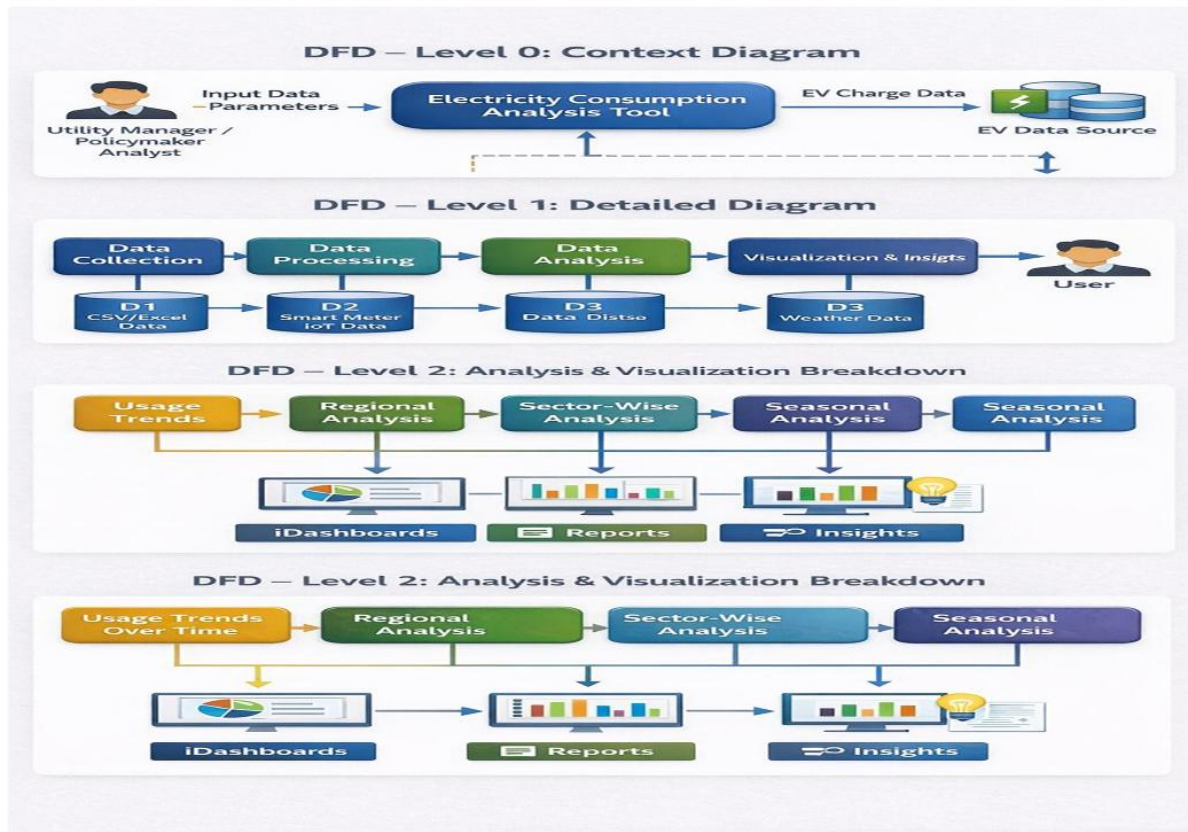
Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand, visually clear, and user-friendly, allowing users to analyze electricity consumption with minimal effort.
NFR-2	Security	The data used for analysis should be stored securely, with controlled access to prevent unauthorized modification or misuse.
NFR-3	Reliability	The system should consistently display accurate electricity consumption data without errors or data loss.
NFR-4	Performance	Dashboards and visualizations should load quickly and respond smoothly to user interactions such as filters and selections.

Following are the non-functional requirements of the proposed solution

NFR-5	Availability	The Tableau dashboards should be accessible whenever required, ensuring continuous availability for analysis and reporting.
NFR-6	Scalability	The system should handle additional datasets, more years of data, and expanded regional information without performance degradation.

2.3 Data Flow Diagram



User Stories

Use the below template to list all the user stories for the product

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Data Access	USN-1	As a user, I want to load electricity consumption data into Tableau so that I can analyze it visually.	Data is successfully imported and visible in Tableau	High	Sprint-1
Customer (Mobile user)	Data Preprocessing	USN-2	As a user, I want to clean and preprocess the data to remove missing or inconsistent values.	Cleaned dataset is ready for analysis	High	Sprint-1
Customer (Mobile user)	Time-based Analysis	USN-3	As a user, I want to analyze electricity consumption trends over time (monthly/yearly).	Line charts show correct time-based trends	High	Sprint-1
Customer (Mobile user)	Regional Analysis	USN-4	As a user, I want to compare electricity consumption across different states or regions.	Bar/Map visuals show region-wise consumption	High	Sprint-2
Customer (Mobile user)	Sector-wise Analysis	USN-5	As a user, I want to analyze electricity usage by residential, commercial, and industrial sectors.	Sector-wise comparison is displayed clearly	Medium	Sprint-2
Customer (Mobile user)	Dashboard	USN-6	As a user, I want to view all key insights in a single interactive dashboard.	Dashboard loads with multiple visualizations	High	Sprint-2
Customer (Mobile user)	Filters & Interaction	USN-7	As a user, I want to filter data by year, state, and sector for better analysis.	Filters update visuals dynamically	Medium	Sprint-3
Customer (Mobile user)	Comparative Analysis	USN-8	As a user, I want to compare electricity consumption between different years (2019 vs 2020).	Comparison charts show accurate differences	High	Sprint-3
Customer (Mobile user)	Storytelling	USN-9	As a user, I want to create a Tableau story to explain insights step-by-step.	Story contains ordered visuals with explanations	Medium	Sprint-3
Customer (Mobile user)	Reporting	USN-10	As an evaluator, I want to review dashboards and insights clearly for assessment.	Dashboards and stories are easy to understand	High	Sprint-4

2.4 Technology Stack

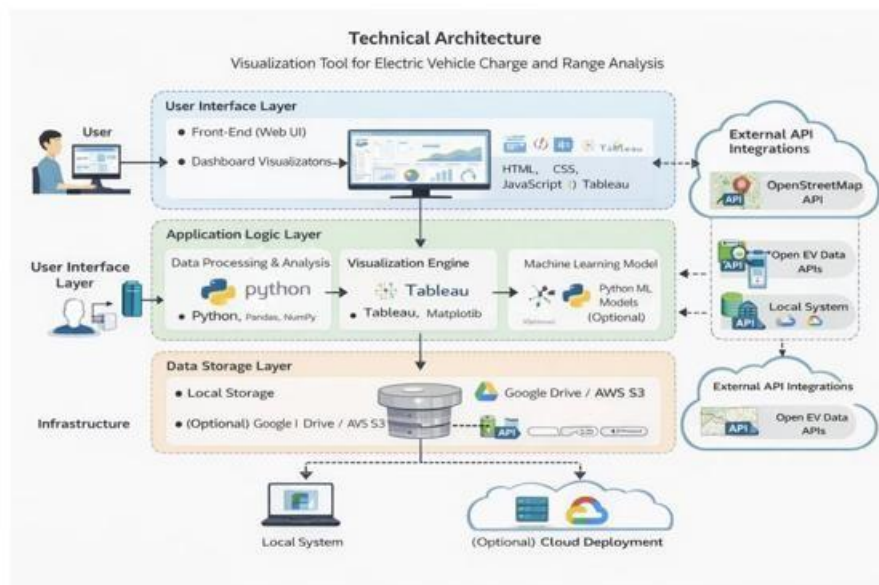


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface to view electric consumption visualizations with Tableau	HTML, CSS, JavaScript, Tableau
2.	Application Logic-1	Data processing and analysis	Python

3.	Application Logic-2	Data cleaning and transformation	Pandas, NumPy
4.	Application Logic-3	Visualization logic	Tableau, Matplotlib
5.	Database	Extracting from consumption file	CSV Files
6.	Cloud Database	Cloud storage for datasets	. Google Drive
7.	File Storage	Storage of reports and datasets	Local File System
8.	External API-1	Electric consumption data statewide	Open Electric consumption Data APIs
9.	External API-2	Geographic data for maps	Open in Tableau
10.	Machine Learning Model	Range estimation (optional)	Python ML Models
11.	Infrastructure (Server / Cloud)	Application deployment	Local System

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for development	Python, Tableau
2.	Security Implementations	Data access control and protection	File permissions, basic encryption
3.	Scalable Architecture	Supports adding more datasets and charts	Modular architecture
4.	Availability	System available during analysis time	Local
5.	Performance	Fast loading dashboards	Optimized queries, caching

3. PROJECT DESIGN

3.1 Problem Solution Fit

Define CS, fit into CC Focus on J&P, tap into BE, understand RC	1. CUSTOMER SEGMENT(S) CS - Urban households - Apartment residents - Small and medium businesses - Educational institutions - Environmentally conscious consumers - Utility companies (secondary segment)	6. CUSTOMER CONSTRAINTS CC - Limited technical knowledge - Budget constraints - Resistance to new technology - Privacy concerns regarding data usage - Compatibility with existing electrical systems	5. AVAILABLE SOLUTIONS AS - Traditional electricity meters - Monthly paper/online billing statements - Basic smart meters (without analytics) - Manual tracking through spreadsheets - Energy-efficient appliances	Explore AS, differen Focus on J&P, tap into BE, understand RC
	2. JOBS-TO-BE-DONE / PROBLEMS CC - Monitor monthly electricity consumption - Reduce high electricity bills - Identify high energy-consuming appliances - Avoid peak-hour overuse - Track and control energy usage remotely - Adopt sustainable energy practices	9. PROBLEM ROOT CAUSE RC - Lack of real-time visibility of electricity usage - Limited awareness of energy-saving practices - No predictive insights on future consumption - Inefficient appliances and unmanaged usage habits	7. BEHAVIOUR BE - Consumers check electricity bill only once a month - Limited monitoring of daily usage - Overuse during peak hours - Reactive approach instead of preventive management - Gradual shift toward smart home devices	

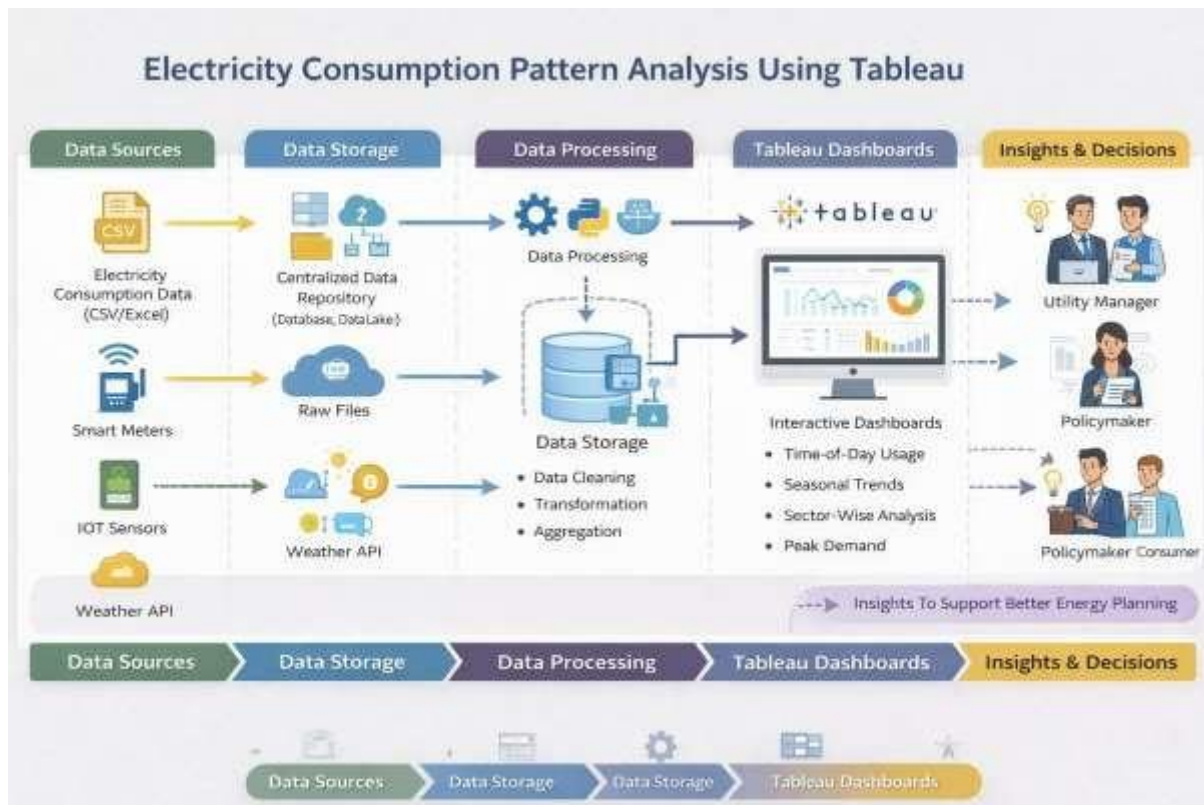
3.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Rapid urbanization, increasing dependence on electronic devices, and inefficient energy usage have led to rising electricity consumption. This results in higher costs for consumers, strain on power grids, and increased carbon emissions. There is a need for a smarter, data-driven approach to monitor, analyze, and optimize electricity consumption at household and community levels.

2.	Idea / Solution description	Develop a Smart Electricity Monitoring and Analytics System that tracks real-time energy usage using IoT-enabled smart meters. The system will provide: • Real-time consumption data via a mobile/web dashboard • AI-based usage pattern analysis • Personalized energy-saving recommendations • Peak load alerts and bill prediction • Integration with renewable energy sources (e.g., solar) The platform empowers users to make informed decisions and reduce unnecessary electricity usage.
3.	Novelty / Uniqueness	• Combines IoT + AI analytics + user-friendly visualization in one integrated platform • Predictive energy consumption forecasting • Gamified energy-saving rewards system • Community-level comparison dashboards to encourage competitive energy saving • Adaptive AI that learns user behavior and suggests automated optimizations
4.	Social Impact / Customer Satisfaction	Reduces electricity bills for households Promotes sustainable energy consumption habits

		Decreases carbon footprint and environmental impact Encourages responsible energy usage in communities Improves customer awareness and control over energy expenses
5.	Business Model (Revenue Model)	Subscription-based model (monthly/annual premium analytics plans) Sale or leasing of smart monitoring hardware Partnerships with utility companies Data insights services for energy providers (anonymized analytics) Commission from renewable energy integration services
6.	Scalability of the Solution	Cloud-based infrastructure allows nationwide and global deployment Modular IoT system adaptable for homes, offices, industries, and smart cities Easy integration with existing smart meters and grid systems Expandable to include water and gas consumption monitoring Suitable for both developed and developing regions

3.3 Solution Architecture



4. PROJECT PLANNING

4.1 Performance Testing

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint 1	Introduction to Electricity	US-01	As a user, I want to understand basic concepts of electricity so that I can build foundational knowledge	3	High	Self
Sprint 1	Sources of Electricity	US-02	As a user, I want to learn about renewable and non-renewable energy sources	5	High	Self
Sprint 2	Electricity Generation	US-03	As a user, I want to explore how electricity is generated in power plants	5	High	self
Sprint 2	Electricity Transmission	US-04	As a user, I want to understand how electricity is transmitted and distributed	3	Medium	self
Sprint 3	Applications of Electricity	US-05	As a user, I want to see real-life applications of electricity in homes and industries	5	High	self
Sprint 3	Future of Electricity	US-06	As a user, I want to learn about future technologies like smart grids and EVs	8	High	self
Sprint 4	Energy Conservation	US-07	As a user, I want to know ways to save electricity and reduce wastage	3	Medium	self
Sprint 4	Final Review & Presentation	US-08	As a user, I want to compile and present the project findings clearly	2	Medium	self

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	2 February 2026	7 February 2026	18	7 February 2026
Sprint-2	20	6 Days	9 February 2026	14 February 2026	19	14 February 2026
Sprint-3	20	6 Days	16 February 2026	21 February 2026	20	21 February 2026
Sprint-4	20	6 Days	23 February 2026	28 February 2026	20	28 February 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Velocity Calculation

Velocity = Total Story Points Completed ÷ Number of Sprints

Total completed story points

$18 + 19 + 20 + 20 = 77$ story points

Number of sprints

= 4

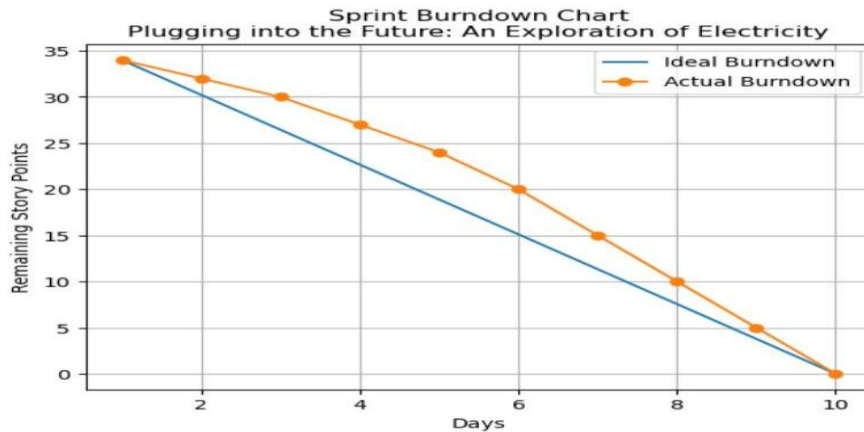
Average Team Velocity:

Average Team Velocity Calculation

Velocity = $77 / 4 = 19.25$ story points per sprint

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time



5. PROJECT DEVELOPMENT PHASE

5.1. Data Connectivity

The data required for this project is collected from publicly available sources and stored in CSV file format. The datasets include electric vehicle specifications and charging station details. These CSV files are directly connected to Tableau for visualization and analysis.

Links of Data Sets:

1. https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view?usp=sharing

5.2. Data Preparation

Data preparation is an important step to ensure accuracy and consistency. The following preprocessing steps were performed before visualization:

- Removed missing and null values
- Removed duplicate records
- Renamed column headers for better understanding
- Converted different dates into a standard format

- Filtered unnecessary attributes
- Checked data consistency and corrected errors

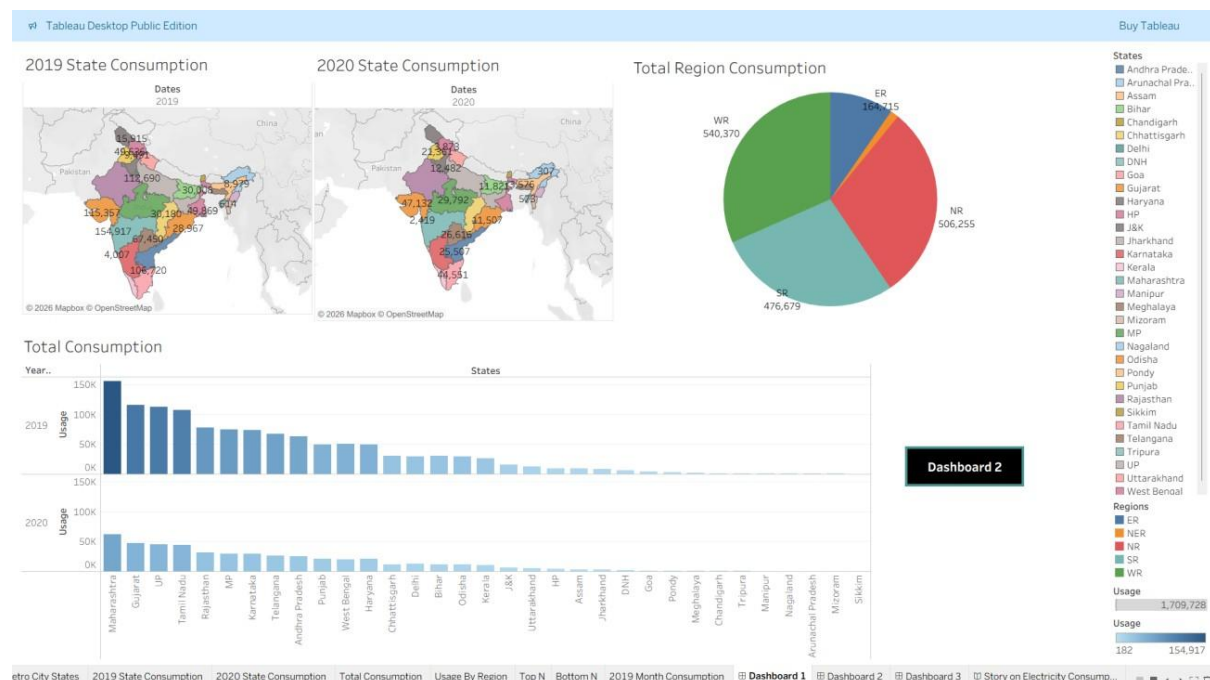
After pre-processing ,the cleaned data was used for creating dashboards and stories.

5.3. Dashboard

A Tableau dashboard was created by combining multiple visualizations to provide a clear understanding of Electric consumption in 2019 and 2020 analysis. The dashboard includes:

- This dashboard helps us compare electricity usage across states for 2019 and 2020, so we can see how consumption changed over time.
- The maps make it easy to visually understand which states use more or less electricity instead of just looking at numbers.
- The region-wise pie chart clearly shows which part of the country consumes the most electricity.
- The bar charts rank states based on consumption, making it quick to identify top and bottom states.

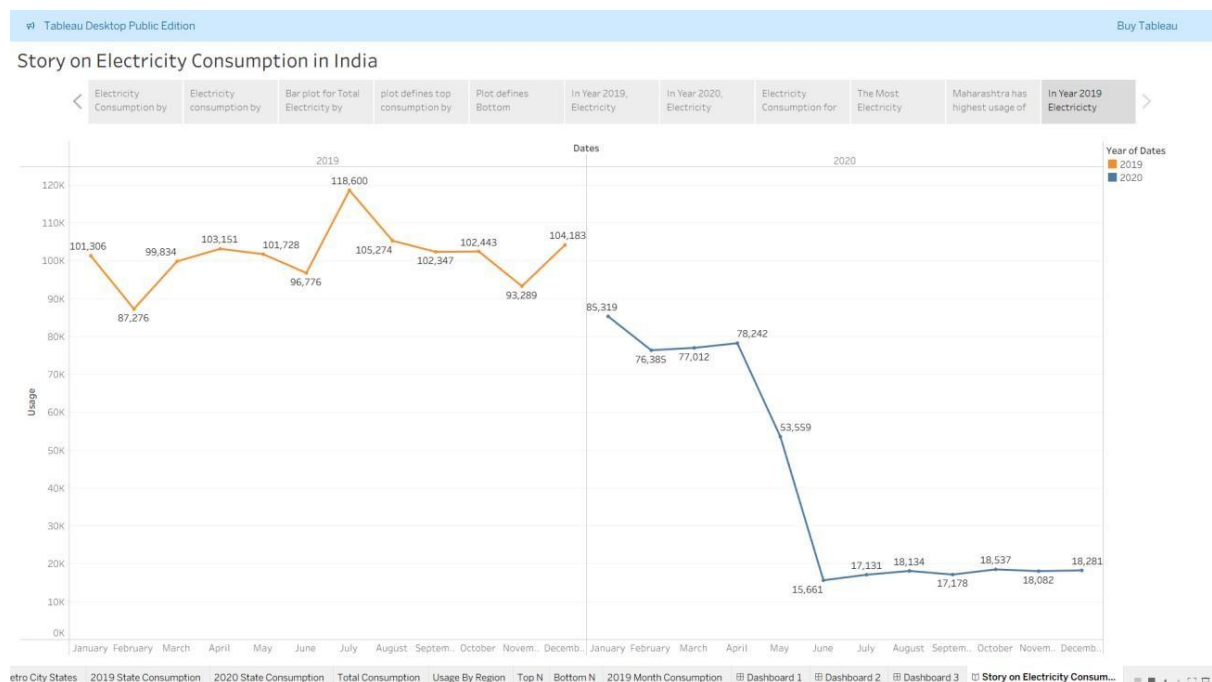
The dashboard is interactive and allows users to filter data based on Electricity Consumption



5.4. Story

A Tableau story was created to present insights in a step-by-step manner. The story consists of multiple story points that guide the user through the analysis:

1. Shows monthly electricity consumption trend for the year 2019.
2. Displays monthly electricity consumption trend for the year 2020.
3. Comparison between electricity usage in 2019 and 2020
4. Most consumption and least consumption in given states
5. Key insights and observations



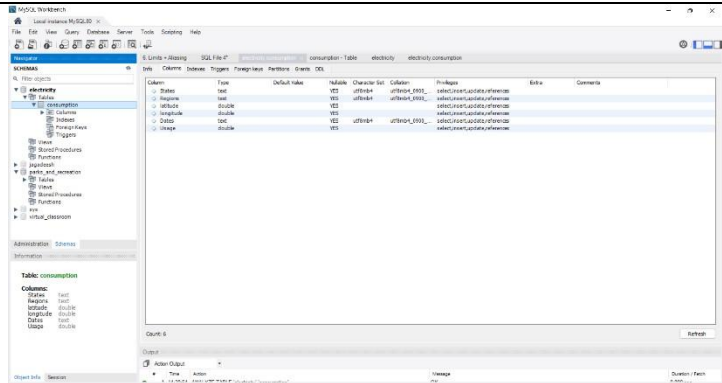
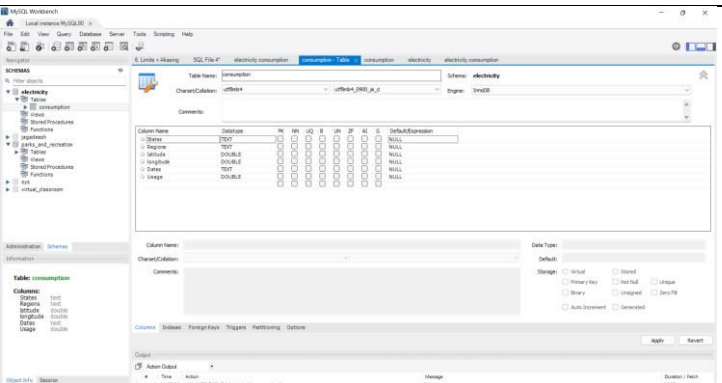
5.5. Creativity(FontandStyle)

The project document is designed using a clean and professional layout. Consistent fonts, proper headings, and structured sections are used throughout the document. Visual elements such as charts and dashboards are clearly labeled and aligned. The overall design improves readability and makes the project easy to understand.

1. Applied clear headings and sub headings to separate sections
2. Used bullet points instead of long paragraphs to improve clarity
3. Maintained proper spacing and alignment across all pages
4. Used meaningful titles and captions for dashboards and stories
5. Applied color the in dashboard story visually differentiate data categories

6. PERFORMANCE TESTING

Project team shall fill the following information in model performance testing template

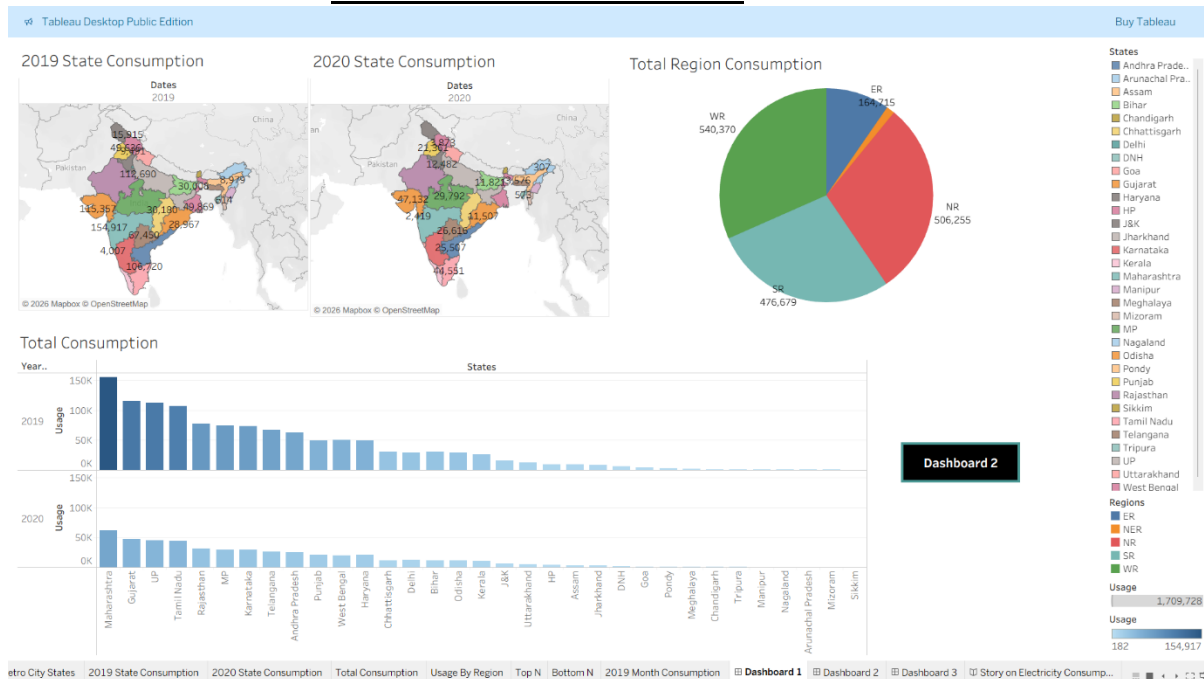
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Preprocessing	
3.	Utilization of Filters	
4.	Calculation fields Used	Total electricity consumption, percentage of electrical consumption records, trends and calculations.
5.	Dashboard design	No of Visualizations / Graphs - 15

6	Story Design	No of Visualizations / Graphs -3
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7. output

Output with screenshots

DASHBOARD SCREENSHOTS



STORY SCREENSHOTS

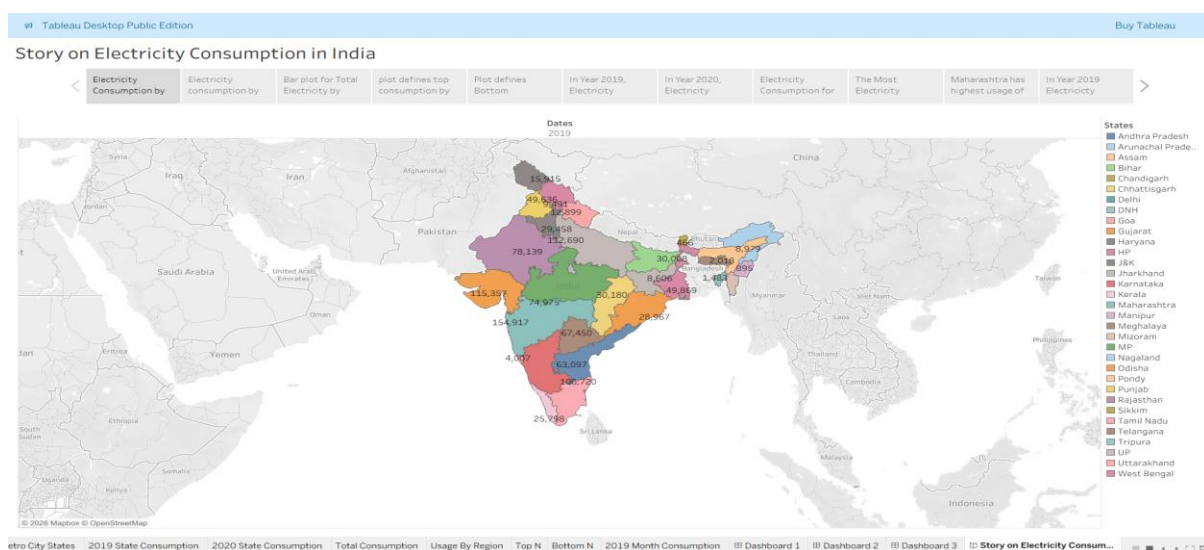


Tableau Desktop Public Edition

Buy Tableau

Story on Electricity Consumption in India

Electricity Consumption by Electricity consumption by Bar plot for Total Electricity by plot defines top consumption by Plot defines Bottom In Year 2019, Electricity In Year 2020, Electricity Electricity Consumption for The Most Electricity Maharashtra has highest usage of In Year 2019 Electricity

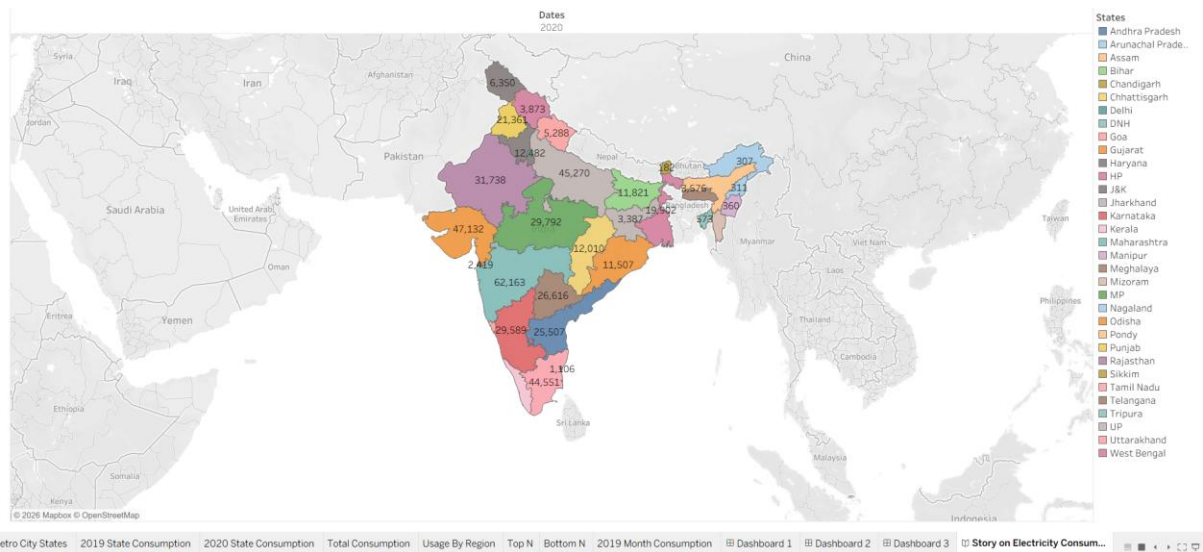


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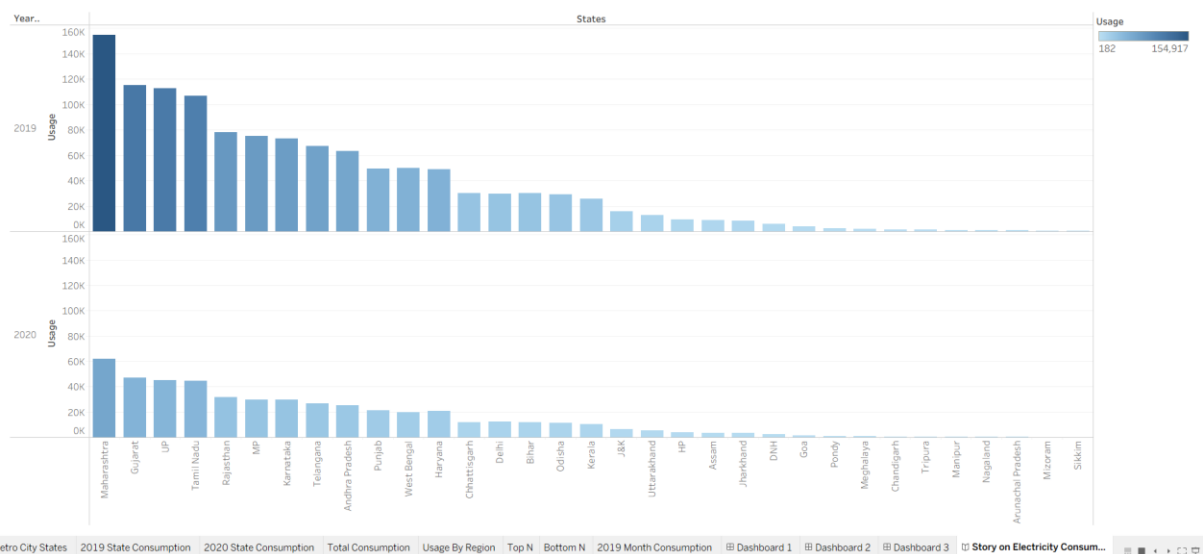


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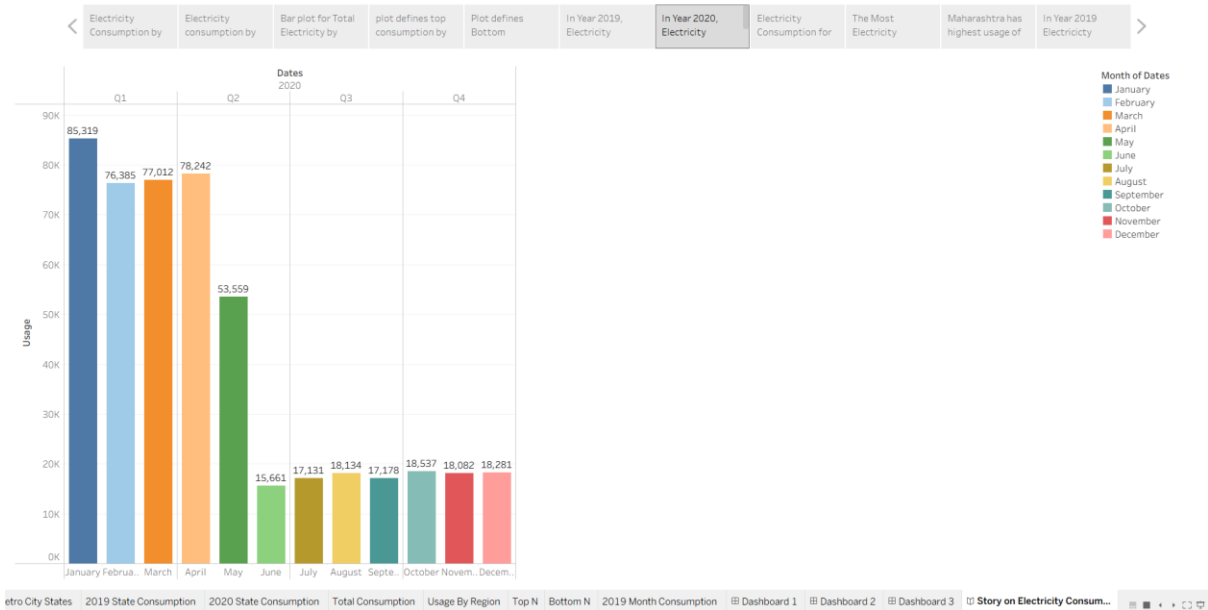


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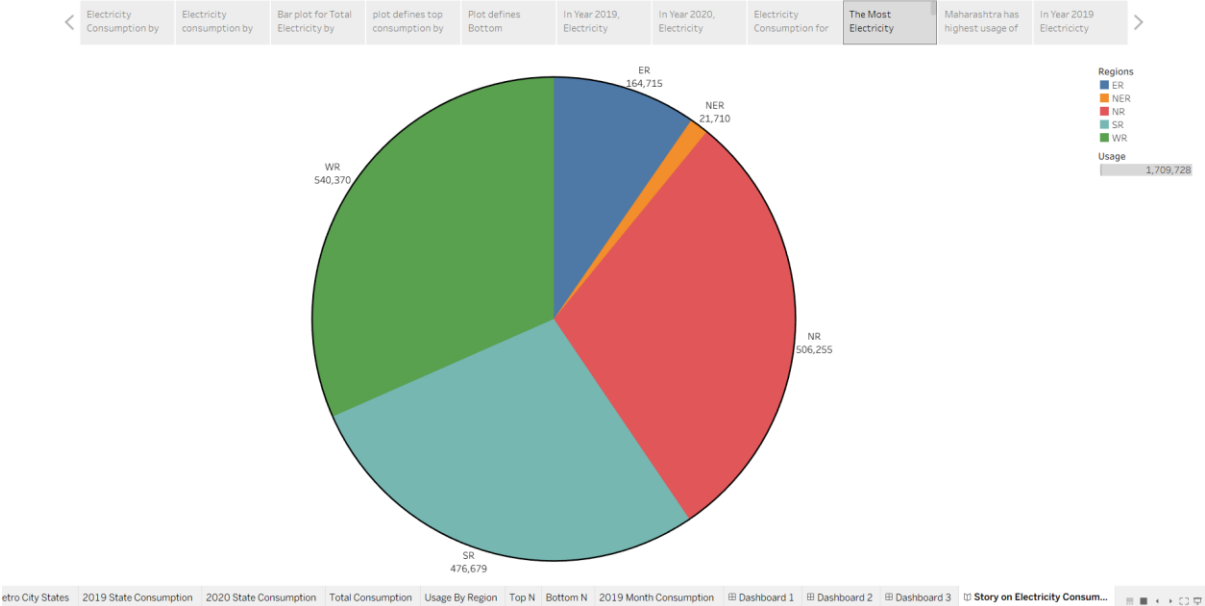
Story on Electricity Consumption in India

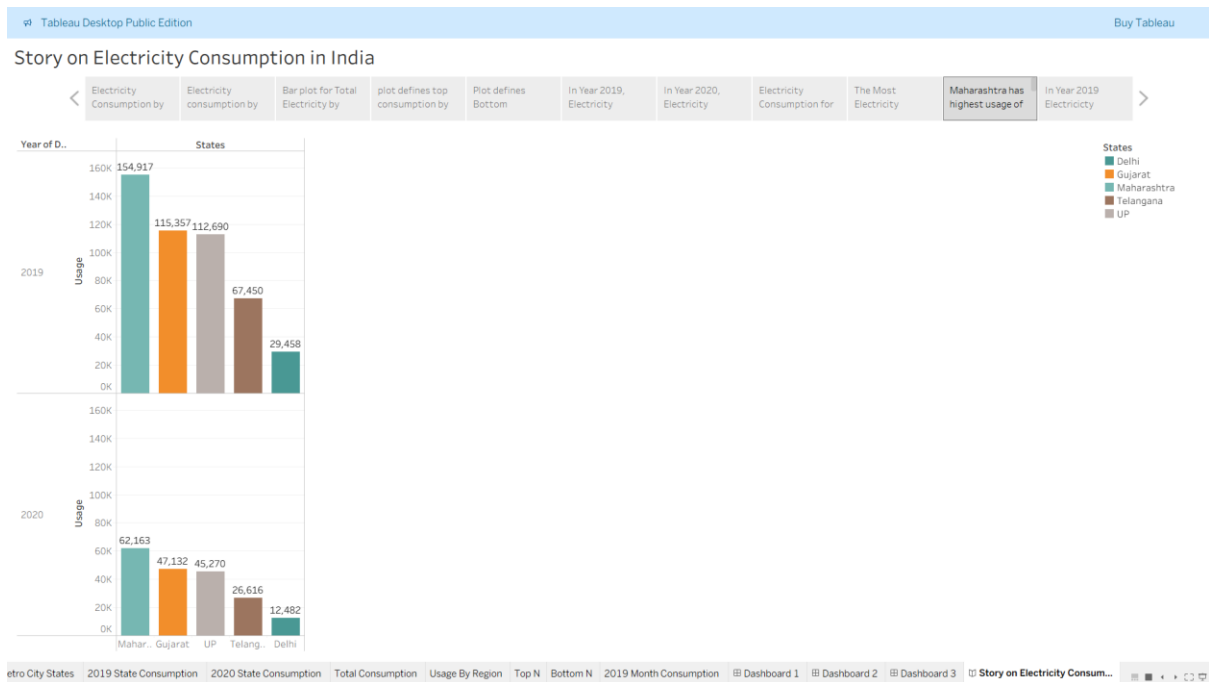


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Story on Electricity Consumption in India





8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

1. Provides interactive and visual insights into electric vehicle charging behavior and driving range efficiency.

2. Helps users make informed decisions by analyzing charging patterns under different conditions.
3. User-friendly dashboards improve understanding of complex Electric consumption analysis performance data.
4. Supports data-driven planning for Electric consumption usage, charging infrastructure, and energy optimization.
5. Can be enhanced with real-time data and predictive analytics for future scalability.

8.2 Disadvantages

1. Accuracy of analysis depends heavily on the quality and availability of input data.
2. Initial setup and integration with external APIs may require technical expertise.
3. Real-time data processing can increase system complexity and resource usage.
4. Visualization tools may have limitations when handling extremely large datasets.
5. Advanced features like machine learning models require additional computational resources.

9. CONCLUSION

This project, An Exploration of Electricity Consumption Patterns using Tableau, successfully analyzes electricity usage trends across different regions and time periods through interactive data visualizations. By leveraging Tableau's powerful analytical capabilities, complex datasets were transformed into intuitive dashboards that clearly highlight consumption patterns, regional variations, and temporal trends.

The visual insights generated through this project support better understanding of electricity demand behavior and aid stakeholders in identifying growth areas, inefficiencies, and usage fluctuations. Overall, the project demonstrates how data visualization tools like Tableau can play a vital role in energy analysis and informed decision-making, with potential for future enhancement through real-time data integration and predictive analytics.

10. FUTURE SCOPE

The scope of An Exploration of Electricity Consumption Patterns using Tableau can be further expanded by integrating real-time electricity consumption data to enable live monitoring and instant insights. Advanced analytics such as forecasting models can be incorporated to predict future electricity demand based on historical trends. The project can also be enhanced by

including weather, population, and industrial data to understand external factors affecting electricity usage. Additionally, deploying the dashboards on Tableau Server or Tableau Public would allow wider accessibility and support data-driven decision-making for policymakers and energy providers.

11. APPENDIX

Source Code(Dataset Links)

https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view?usp=sharing

GitHub & Project Demo Link

TEAM LEADER AND MEMBER GITHUB LINK :

Link of Github :

<https://github.com/Jagadish2005-tech/Plugging-into-the-Future-An-Exploration-of-Electricity-Consumption-Patterns-Using-Tableau.git>

PROJECT LINK

Link of the video :

<https://drive.google.com/file/d/1EDUKq93CQmh8bQH97F1nrhnxGEBnboyC/view?usp=sharing>

